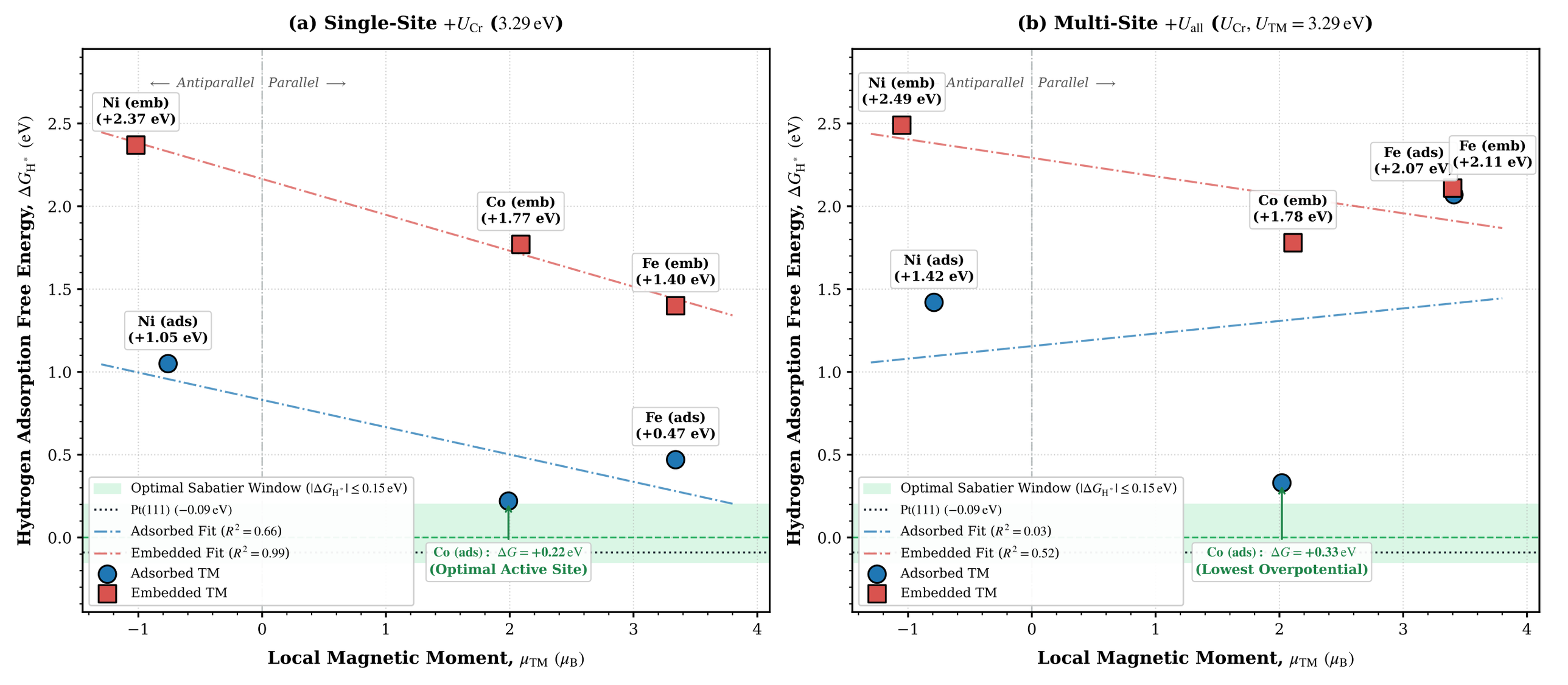

(a) Single-Site +U_{Cr} (3.29 eV)
(b) Multi-Site +U_{all} (U_{Cr}, U_{TM} = 3.29 eV)
Hydrogen Adsorption Free Energy, ΔG_{H^*} (eV)
Local Magnetic Moment, μ_{TM} (μ_B)
⟵ Antiparallel Parallel ⟶
Ni (emb) (+2.37 eV)
Co (emb) (+1.77 eV)
Fe (emb) (+1.40 eV)
Ni (ads) (+1.05 eV)
Fe (ads) (+0.47 eV)
Co (ads): ΔG = +0.22 eV
(Optimal Active Site)
Optimal Sabatier Window (|ΔG_H*| ≤ 0.15 eV)
Pt(111) (−0.09 eV)
Adsorbed Fit (R^2 = 0.66)
Embedded Fit (R^2 = 0.99)
Adsorbed TM
Embedded TM
Ni (emb) (+2.49 eV)
Co (emb) (+1.78 eV)
Fe (ads) (+2.07 eV)
Fe (emb) (+2.11 eV)
Ni (ads) (+1.42 eV)
Co (ads): ΔG = +0.33 eV
(Lowest Overpotential)
Adsorbed Fit (R^2 = 0.03)
Embedded Fit (R^2 = 0.52)